Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs in the Age of Transparent AI

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Problem Statement

The Core Question: How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2, RMSE)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability, statistical inference)

Research Objectives:

1. Compare ML and Statistical approaches across multiple problem domains
2. Quantify trade-offs between performance, cost, and interpretability
3. Provide evidence-based recommendations for model selection
4. Connect findings to Interpretable ML research and Transparent AI

Scope of the Study

Category	Count	
Problem Domains	4	Regression,
Datasets	13	
ML Models	5	Linear Regression, Random F
Statistical Models	5	OLS Regression, Logist
Total Comparisons	13	

Problem Domains & Datasets:

- **Regression:** California Housing, Diabetes, Wine Quality, Energy Consumption
- **Classification:** Iris, Wine Classification, Breast Cancer, Credit Default
- **Time Series:** Energy Consumption, Temperature, Sales Forecasting
- **Factorial Analysis:** Treatment Effects, Educational Outcomes, Agricultural Experiment, Drug Efficacy Study

Methodology: Regression

Statistical:

- OLS Regression (statsmodels)
- Provides p-values, confidence intervals
- Statistical inference capabilities

ML:

- Linear Regression (scikit-learn)
- Optimized for prediction
- Fast training and inference

Results:

Dataset	OLS R^2	ML LR R^2
California Housing	0.575	0.575
Diabetes	0.453	0.453
Wine Quality	0.787	0.787
Energy Consumption	0.811	0.970

Methodology: Classification

Statistical:

- Logistic Regression (statsmodels)
- Maximum Likelihood Estimation
- Provides odds ratios, p-values

ML:

- Random Forest (scikit-learn)
- Ensemble of decision trees
- Feature importance scores

Results:

Dataset	Logistic Acc.	RF Acc.
Iris	1.000	1.000
Wine Classification	0.956	0.982
Breast Cancer	0.956	0.965
Credit Default	0.999	0.999

Methodology: Factorial Analysis

Statistical:

- ANOVA (Analysis of Variance)
- Tests null hypothesis
- Provides F-statistics and p-values

ML:

- Support Vector Regression (SVM)
- RBF kernel for non-linear relationships
- Hyperparameter tuning via grid search

Results:

Dataset	ANOVA R^2	SVM R^2
Treatment Effects	0.417	0.513
Educational Outcomes	-0.002	0.284
Agricultural Experiment	0.464	0.745
Drug Efficacy Study	0.728	0.880

Methodology: Time Series

Statistical:

- ARIMA (AutoRegressive Integrated Moving Average)
- Auto-selection of (p, d, q)
- Interpretable coefficients
- Statistical diagnostics

Deep Learning:

- RNN/LSTM (PyTorch)
- Captures long-term dependencies
- Handles complex temporal patterns
- Requires more data and computation

Results:

Dataset	ARIMA R^2	LSTM R^2
Energy Consumption	0.450	0.890
Temperature	0.972	0.965
Sales Forecasting	0.972	0.969

Methodology: Survival Analysis

Statistical:

- Cox Regression (lifelines)
- Poisson Regression (statsmodels)
- Purpose-built for censored data
- Interpretable hazard ratios

ML:

- Gradient Boosting (scikit-learn)
- Handles complex patterns
- Requires adaptation for censoring

Results:

Dataset	GBM R^2	Cox R^2	Poisson R^2
Patient Survival	0.108	-0.182	0.340
Customer Churn	-0.190	-3.923	-0.022
Equipment Failure	-0.263	-0.814	0.084

Key Finding: Poisson Regression outperforms both Cox and GBM, with 40-400x faster training.

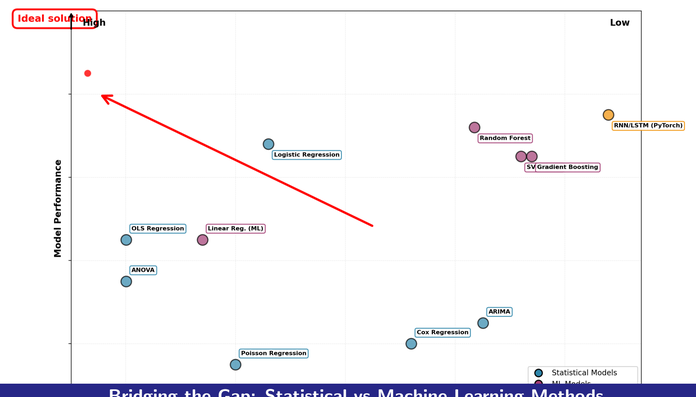
Overall Performance Summary

Metric	Statistical	ML/DL
Avg Performance (R^2/Acc)	0.547	0.818
Avg Training Time	0.037s	0.763s
Efficiency (Perf/Cost)	High	Medium
Interpretability	88.3/100	37.4/100

Key Statistics:

- Statistical models: 20-100x faster training
- ML/DL models: 1.5x higher average performance
- Interpretability gap: 2.4x difference

Performance vs Training Cost



The Fundamental Trade-offs

$P(\theta) = \mathbb{E}[\mathcal{M}(Y, \hat{Y})]$ — Metrics: R^2, Accuracy, RMSE

$C(\theta) = T_{\text{train}}(\theta) + \lambda M(\theta)$ — Cost: Training time + memory cost

$I(\theta) = \text{Expl}$ — Interpreta: Transparenc trust

The Optimization Problem:

$$\max_{\theta} \quad P(\theta) - \alpha C(\theta) + \beta I(\theta)$$

s.t. constraints

Efficiency Analysis

Ranking	Model	Efficiency
1	OLS Regression (Stat)	6435.64
2	ANOVA (Stat)	5445.54
3	Linear Regression (ML)	1297.41
4	Logistic Regression (Stat)	439.78
5	Poisson Regression (Stat)	349.65
...	...	...
Last	RNN/LSTM (DL)	0.36

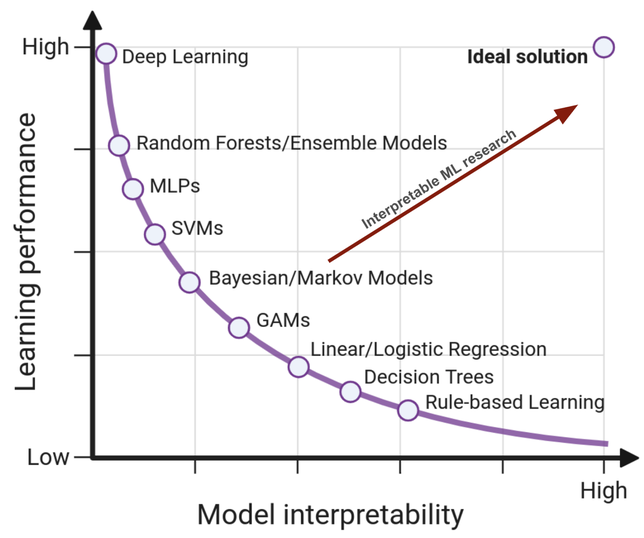
Statistical Models:

- 20-100x more efficient
- Best for real-time applications
- Low computational cost

ML/DL Models:

- Higher performance potential
- Better for complex patterns
- Suitable for batch processing

Performance vs Interpretability Trade-off



The Interpretability-Performance Curve:

- Traditional trade-off: Higher performance ⇔ Lower interpretability
- **Interpretable ML Research:** Moving toward the ideal solution
- Goal: Bridge the gap between statistical and ML approaches

When to Choose Statistical Models

Choose Statistical Models When:

- ✓ Interpretability is critical - Need p-values, confidence intervals
- ✓ Regulatory compliance - Transparent decision-making required
- ✓ Hypothesis testing - Understanding causal relationships
- ✓ Small to medium datasets - Optimal performance on limited data
- ✓ Real-time applications - Fast training and prediction needed
- ✓ Research and development - Need to understand underlying relationships

Best Models: OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox Regression

When to Choose ML/DL Models

Choose ML/DL Models When:

- ✓ Maximum accuracy is priority - Need best predictive performance
- ✓ Large, complex datasets - High-dimensional or non-linear patterns
- ✓ Production systems - Computational resources available